2 Pair Programming

Pair programming means that you create the code together with an artificial intelligence (AI) tool. It's rarely a good idea to prompt the AI tool with statements like, "My task is such and such, give me the solution!" This only works for simple problems or relatively short functions that are self-contained.

Most of the chapters in this book are intended for programmers or developers who are already reasonably proficient in their craft. We therefore assume that you're familiar with at least one programming language and know how to use functions, methods, and classes. Ideally, you also have some IT background knowledge, for example, you can work with databases or Git.

We make an exception to this rule in this chapter because here we specifically address those of you who are in the process of learning a programming language. In the examples, we demonstrate how AI tools can help you.

But although we focus on the learning process in this chapter, pair programming is, of course, also suitable for professionals! The greater your prior knowledge and the more complex your project, the more important it is that you proceed step-by-step; that is, specify the structure of the program and solve or improve one detail at a time with the help of AI. So, the aim of this chapter isn't to show you how an international bank account number (IBAN) verification function or a sudoku solver work. Rather, these sample applications are intended to give you examples of a reasonable use of AI tools. We want to show you what prompt engineering looks like in everyday work.

Our examples use the widely used Python and Java languages. But if you're learning another language, it doesn't matter! The prompts presented here work just as well in other languages. As in the other chapters of this book, it's not primarily about the answers provided by the AI tool (which we've often not presented at all or only in abbreviated form), but about how you can ask the AI tool questions in the most effective way.

The Ideal AI System

We've used ChatGPT with GPT-4o throughout for the examples in this chapter. However, you can just as easily use one of the competing chat systems. We had a particularly good experience using Claude from Anthropic while working on this book (**https://claude.ai/chat**). At the time of our tests in early 2025, Claude not only provided excellent quality answers but was also a little less verbose. The shorter answers are often easier to digest, especially if you only have a modest knowledge of programming. Even though ChatGPT is currently the best known, you can see there are exciting alternatives that—when not used extensively—are also free of charge.

If you're learning to program and don't want to complete a task as quickly as possible, then we strongly advise against using a coding wizard such as GitHub Copilot. Typical exercises such as "Write a function that tests whether the parameter is a prime number" or "Develop a class to map a bank account" are trivial for code wizards. All you have to do is enter a plausible function name such as `testIfPrime`, and the AI assistant will return the entire code. That means you'll finish quickly, but learn little.

Chat-based systems are better suited for learning. Ideally, you should explicitly state in the first prompt of a chat history that you're a beginner who needs some help and food for thought, but no finished code.

2.1 Structuring Code into Functions

Even if we're not assuming years of coding practice in this chapter, we're not starting from scratch either. There are other books for that. We assume that you already know what variables, branches, loops, and so on are and that you can handle these language constructs.

Experience shows that with this level of knowledge, the next hurdle is to structure longer code into functions in a meaningful way. That is the point at which we start here.

One starting point is to simply hand your AI chat partner the (spaghetti) code you've written and ask whether it sees a way of splitting the code across multiple functions.

Prompt: I wrote this Python code to estimate the quality of a password. Can you help me break the code down into functions? Please explain why and how.

[50 lines of code]

We can tell you this much: The code will almost always look neater afterward. ChatGPT also suggests meaningful function and parameter names and takes care of the documentation of the code in case you've neglected it. Try the preceding prompt or a similar formulation with your own code! (By the way, we go into more detail about improving existing code with AI help in **Chapter 4**.)

2.1.1 Defining Functions Yourself

At first, the procedure outlined earlier is instructive, but sooner or later, you should be able to divide your code into subfunctions yourself. As long as you find this difficult, you can ask for specific help.

Note the structure of the prompt: It begins with a relatively detailed summary of background information with regard to which programming language, which objectives, and which data structures you used. In addition, you should try to make it clear to your AI chat partner that it shouldn't spit out the finished code straight away (which would be easy for it), but that you want explicit support in structuring the code.

Prompt: In my Python script, I have a nested list of numbers. I need to test if this list is a "magic square" where the row, column, and diagonal sums are all equal. Please help me *only* find a structure for the code. How should

I define and use functions? Please also explain parameters. Don't provide the code; I want to write it myself.

> To structure the code for checking if a nested list of numbers is a magic square, you can break down the problem into several smaller functions. Here's how you can define and use functions, along with an explanation of their parameters:
>
> - Function to Check Square Shape:
>   - Function Name: `is_square`
>   - Parameters: `matrix` (a nested list of numbers)
>   - Purpose: Ensure that the input is a square matrix (i.e., it has the same number of rows and columns).
> - Function to Calculate Row Sums (. . .)
> - Function to Calculate Column Sums (. . .)
> - Function to Calculate Diagonal Sums (. . .)
> - Function to Check Uniform Sums (. . .)
> - Main Function to Check Magic Square (. . .)
>
> [more details]

The answer is an excellent starting point for implementing the program step by step.

2.1.2 Defining Functions and Parameters

The starting point for the second example is an exercise. You're supposed to program a function that removes duplicates from a list. Your first problem isn't so much the code itself, but rather the definition of the function and the resulting data flow. Obviously, you'll pass the list to be processed as a parameter to the function. But how should the function return the result?

In the prompt, you again summarize the task and specify which programming language you're using. Again, what is important here is that you're not interested in the code, but in assistance with the declaration of the function and the structure of your code.

Prompt: I need to write a Python function to remove duplicates from a list. How should I define the function? Don't provide actual code for the function.

> To define a function for removing duplicates from a list in Python, you can follow these steps to structure the function:
>
> - Use a clear and descriptive name, such as `remove_duplicates`.
> - Parameters: Only `lst`, the list from which you want to remove duplicates.
> - Return Value: The function should return a new list with duplicates removed.
> - Method to Remove Duplicates: You can use different methods to remove duplicates, such as using a set or iterating through the list and checking for existing elements. Consider how you want to handle the order of elements (preserve or not).
> - Edge Cases: Consider edge cases such as an empty list or a list with all identical elements.
>
> [detailed explanation with more details]

Following the preceding, already shortened answer, ChatGPT can't resist the temptation to deliver (almost) finished code after all. Aside from that, however, the answer is a good basis for writing your own code. We particularly liked the fact that the AI tool points out that your function also has to cope with special cases (empty lists).

Perhaps you've already used other programming languages in the past. For some languages, you can change the data transferred in the parameter. This can increase the efficiency of your code, especially with large amounts of data. What does that look like in Python?

Prompt: Instead of returning the result, is it possible or even recommended to modify the data passed as a parameter to a function?

> Yes, it's possible to modify the data passed as a parameter to a function in Python, especially if the data type is mutable, such as a list. However,

> whether it's recommended depends on the context and the specific use case. Here's a detailed explanation.
>
> [two pages of details]

The answer is symptomatic: ChatGPT readily generates endless amounts of text to explore the advantages and disadvantages of different variants. In most cases, however, the AI tool avoids making a clear recommendation. In this case, almost everything speaks against a direct modification of the data, especially the readability of the code. Instead, it's better to follow the simple approach of processing parameters as an input and using a return to provide the result (output).

Depending on which language model you use, the answer to the same question may be completely different. The following lines were generated by the free Llama 3 language model, which we executed locally with Ollama. We liked this answer better, both in terms of content and commitment.

> In Python, it's generally not recommended to modify the original data structure.
>
> [more details]
>
> So, while it's technically possible to modify the input list, I would recommend against it.

We also had a go with the AI chat tool Claude from Anthropic. Its answer was similarly noncommittal as the one from ChatGPT, but contained a new aspect: Claude points out that it's sometimes useful to develop *two* versions of a function, one to process the data in place and the second to return results.

> [pros and cons of both approaches]
>
> Alternative approach: A common Python convention is to provide both versions:

> - An in-place function (e.g., `list.sort()`)
> - A function that returns a new object (e.g., `sorted(list)`)

However, the object-oriented approach, that is, the application of a method to a list, is aimed at fairly advanced Python fans.

2.1.3 Detailed Questions about Functions and Parameters

When dealing with functions and parameters, there are a number of special cases that vary greatly depending on the programming language. Of course, you can read the online documentation or a Python book, but you can also simply ask an AI tool. If you don't feel like reading lengthy digressions, you should ask for a concise formulation of the answer. ChatGPT in particular has a tendency to be verbose in its responses and to provide much more detail than you need.

Don't forget to include the name of the programming language you're using in the prompt. (You don't have to do this for follow-up questions, in which case, the language model already knows the correct context.)

The first question refers to the frequently occurring problem that a function should return a multipart result. Note the prompt again: to make it clear to the AI tool what we're talking about, we've also provided a simple example.

Prompt: Can my Python functions return more than one data item (for example, a string and a number)?

> Yes, your Python functions can return more than one data item, such as a string and a number, using several methods. Here are the most common ways:

```
# use a comma-separated list of values to create a
tuple
def get_info():
    return "Hello", 42
# return a list containing the values
```

```python
def get_info():
    return ["Hello", 42]
```

> [three more variants: dictionary, `namedtuple`, custom object]

There are just as many questions for programming beginners about declaring the parameter list of functions. You can ask the AI tool of your choice about this like a fellow student or a tutor.

Prompt: Using Python, how can I define a function to pass a variable number of parameters? And how can I process those parameters inside the function?

Prompt: I want to use optional parameters with default values. What's the Python way to do this?

Prompt: How should I name functions and parameters?

Prompt: Within a Python function, can I access/change variables defined outside?

Prompt: What is the recommended order for a function with several parameters?

ChatGPT is at its best when it comes to such questions and describes every conceivable variant and special case for pages on end. If you don't have the time or inclination to read through this, you need to make it clear to the AI tool that you're only interested in the basics and that you're still at the beginning of programming.

Prompt: Once again, please. Be concise, drop special cases. I am a Python beginner.

You'll get the best answers if you not only ask detailed questions but also provide a specific use case or example.

Prompt: OK. Now, consider only required parameters. What should come first? Suppose I want to write a function to find and replace items in a list.

What's better, `replace(data, find, replace)` or `replace(find, replace, data)`?

> For functions with only required parameters, the general guideline is to order parameters from the most important or the most specific to the least important or the most general [more details]. For a function that replaces items in a list, it's clearer and more intuitive to place the data last:
>
> ```
> def replace(find, replace, data):
>     # Code to replace items
>     pass
> ```
>
> Why this order? [more details]

2.1.4 Code Optimization

You may already have working code that you just want to optimize. ChatGPT can also help you with this. However, be sure to check after each step whether your program is still working as intended!

The first two prompts are about adapting existing code to the usual Python rules. ChatGPT already knows the conventions such as correct indentation and the naming of variables and functions. You only need to go into these details if you have special requests. Ideally, you should then provide templates and specific statements.

Prompt: Please adapt this code to the Python naming conventions.

[100 lines of code]

Prompt: Could you find better names for the variables and functions used in this code?

Good examples are: `total_sales`, `current_temperature`, `user_input_calculcated`.

Don't use: `TotalSales`, `CurrentTemperature`, `UserInputCalculated`.

Many programmers are quite careless when it comes to commenting code. AI tools can also take over this unpopular work (see also **Chapter 6**). In the simplest case, a mini prompt as in the first example is sufficient.

Prompt: Please add comments to this code.

Depending on who is to read your code and comments, you should specify this target group. Note, however, that although the AI tool can usually understand your code, it can't guess *why* you've decided on a particular course of action. AI comments can therefore only partially replace your own comments.

Prompt: Add comments to this file to make the code understandable to a novice programmer.

Tip

Commenting code afterwards can also help if you've received code from a colleague, from the internet, or from another source and don't understand it properly.

Perhaps the function you've developed works but is very slow or takes up a lot of memory. If you ask an AI tool to optimize your code, you should be as specific as possible about what needs to be optimized (e.g., speed).

Prompt: This function is rather slow. Do you see any way to make the code more efficient?

Prompt: When processing large files, this function uses far too much memory. Is there a way to reduce the memory consumption of the code?

2.1.5 Code Reuse

AI tools shouldn't be your only aid when learning a programming language. If you're searching for code on the internet and come across constructions that you don't understand, you can have them explained to you. (Just as you should never blindly accept code generated by the AI unless you really understand it, this rule also applies to code from Stack Overflow or other reputable sites. The code may actually be correct but is intended for a completely different application than the one you have in mind.)

The following prompt shows how you can combine the strengths of the internet as a universal advisor with those of AI. Here, the AI tool is supposed to explain the `linalg.norm` function, which was previously unknown to you.

Prompt: This function apparently calculates the distance between some points. Can you explain what `np.linalg.norm()` does?

```
def get_distances(points_xy):
    l1 = np.linalg.norm(
      points_xy[0, :] - points_xy[1, :])
    l2 = np.linalg.norm(
      points_xy[1, :] - points_xy[2, :])
    return l1, l2
```

You may be switching from one language to another, or you may have found a solution to your problem on the internet in another language. Most AI tools are able to translate your code from one language to another.

Prompt: Last year I wrote a PowerShell script to download the image of a web page. How can I rewrite this code using Python?

[20 lines of PowerShell code]

In our experience, such code conversions don't always work perfectly. Every programming language, every library has its own peculiarities. Take the time to test the code thoroughly and scrutinize its functions. If you want to learn programming and the focus isn't on completing a task quickly, it's even better to use the modules or libraries suggested by the AI bot as a starting point for your own research and then develop the code yourself.

2.1.6 Splitting Code across Multiple Files

For smaller programs, splitting the code into independent functions is sufficient to ensure that the code remains readable, clear, and maintainable.

With complex applications, however, there comes a point where it's advisable to distribute the code across multiple files.

In the following prompt, we asked the AI tool to break down a 250-line code file into subfunctions. We weren't only interested in the completion of the work itself, we also wanted to know why the AI tool chooses a certain approach.

Prompt: I have written a long Python script, and I want to split the code in a logical way across more than one code file. Can you help? Please explain why and how.

[250 lines of code]

> To split your Python script logically across multiple files, we'll use a modular approach. The goal is to organize your code into coherent units that make it easier to maintain, understand, and test. Here's a step-by-step breakdown of how to do it:
>
> - Identify Logical Units [more details]
> - Create Modules [more details]
>   - This is the new code structure:
>
> ```
> project_directory/
> |-- __init__.py
> |-- config.py
> |-- logger.py
> |-- auth.py
> |-- web_utils.py
> |-- main.py
> ```
>
> - Split the Code [more details]
> - Import the Modules: In `main.py`, you import the necessary functions from each module to keep the script organized and modular.
>
> Benefits of modularization [more details]

In the sample code we used, ChatGPT has actually found a sensible way of distributing the code across multiple files, each of which can be easily extended. But perhaps you're not yet familiar with the concept of Python modules. You've probably already used ready-made modules, but haven't yet defined your own.

Prompt: Looks good. However, I am not yet familiar with creating modules. I have used Python modules before, but I have never written modules myself. Can you explain the basic concepts of how to create my own modules? Be concise!

The answer leaves nothing to be desired. However, ChatGPT immediately goes into packages as well. If you don't know what this is (in principle: multiple modules in one directory), you should ask a follow-up question.

Prompt: What's the difference between a module and a package?

If you now take the time to try out the module and package examples suggested by ChatGPT yourself and perhaps expand them here and there, you'll be familiar with the concept an hour or two later. Great!

Tips

There are major differences between common programming languages when it comes to organizing code across multiple files, so—one more time—be sure to specify the language in which you're programming!

Also note that many AI tools have problems processing very long code. While splitting 250 lines of code across over three or four files works wonderfully, ChatGPT and others often fail if the code size exceeds 500 or 1,000 lines. So, don't wait too long before disassembling your code! The reason for the failure with long code files are limits for the size of the context window and for the length of the response (see also **Table 1.1** in **Section 1.2**).